

Zuyu Liu

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Chemical Engineer building at the intersection of **engineering, entrepreneurship, and AI frontiers** to drive industry innovation.

PROFESSIONAL EXPERIENCE

Quality Assurance Intern <i>Procter & Gamble, Tabler Station</i>	Inwood, WV 5/2026 – 8/2026
- Built and piloted a Controlled State system for Beauty Packing line at P&G’s largest global manufacturing site, defining implementation roadmap for further error reduction and process automation with \$4M+ in projected savings. - Led annual Quality risk assessment for mis-pack, mis-code, and mis-label detection sensors for Beauty Packing; coordinated 20+ stakeholders across 4 different teams to deliver gap analysis and mitigation strategies for critical quality controls. - Designed, executed, and validated a new SOP extending shelf life for a critical body wash ingredient, reducing expired-material scrap costs by \$300K+ annually and increasing supply chain flexibility. - Built an AI agent to streamline all site validation document approvals, targeting 30% reduction in manual review effort, 1,600+ productivity hours returned, and \$130K+ in projected annual savings for Quality Department. - Designed an agentic AI workflow for floor-level managers to identify team-specific use cases, drive bottom-up AI adoption, and create \$3M+ in projected annual productivity savings site-wide in partnership with Plant Management.	
Project Manager, prev. Senior Consultant <i>Illinois Business Consulting</i>	Urbana, IL 9/2025 – Present
- Conducted early-stage M&A screening for a Fortune 500 manufacturing company, evaluating acquisition targets across market opportunities, technical capability, and strategic fit. - Led agentic AI workflow design for a Big Four consulting company, reducing a fragmented, multi-month corporate innovation workflow into a structured under-5-minute process using n8n; defined specific AI agents, automation logic, and human-in-the-loop checkpoints.	
Undergraduate Researcher <i>University of Illinois at Urbana-Champaign</i>	Urbana, IL 9/2024 – Present
- Created Sankey Diagrams mapping supply chains for high-performance polymers like PET; visualized feed stock flows, supply-chain gaps, highlighted where sustainable BTX alternatives could create economic and sustainable value, delivered supply-chain gap analysis report. - Conducted computational catalysis research using ASE/VASP (73k CPU-hours), screening 40+ aMOC surface configurations to assess structural stability for hydrogen fuel cell applications.	
Techno-Economic Analyst <i>Argonne National Laboratory</i>	Lemont, IL 9/2023 - 4/2024
Delivered a stacked-electrode Capacitive De-Ionization design and techno-economic analysis for CDI separation for biofuels, translating lab data plus 10+ studies into a feasibility and scalability report.	

ENTREPRENEURSHIP PROJECTS

President; Formerly Head of Events <i>Founders: Illinois Entrepreneurs</i>	Urbana, IL 12/2024 - Present
- Lead the premier entrepreneurship organization at UIUC. Organized Forge, the largest student-run startup weekend in the Midwest (25+ startups and 30+ investors/mentors), and Keywords, the largest AI-hackathon in UIUC history with Y Combinator (400+ participants). - Created CoLini, a student-startup matching platform connecting 200+ active UIUC students with 30+ student startups to accelerate recruiting, team formation, and early growth.	
Founding Team – Strategy and Sustainability <i>LoopEra, Fashion-Tech Startup</i>	Chicago, IL 6/2025 – 8/2025
Planned Chicago’s first Fashion Tech Summit, convening 210+ participants across fashion, AI, software, materials, sustainability, and entrepreneurship; conducted life-cycle analyses of textile fibers to position LoopEra around low-carbon, ethical supply chains.	
Founding Team – Core Chemical Engineer <i>Credence Cleantech Solutions, Clean-Energy Startup</i>	Urbana, IL 4/2025 – 8/2025
Authored two technical whitepapers on Concentrated Solar Power systems by consolidating industry research, competitor insights, and system-level trade-offs to define Credence’s technical direction and unique advantages. Sourced investment leads for \$1.5 million round.	

EDUCATION

<i>B.S. Chemical Engineering, University of Illinois at Urbana-Champaign (GPA: 3.69/4.00)</i>	Expected Graduation: May 2028
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AWARDS & DISTINCTIONS

- Y Combinator Startup School, Summer 2026** (top 5% of applicants)
- U.S. Presidential Scholar** (1 of 161 selected nationally out of 3.8M U.S. high school graduates)
- Chancellor’s Scholar** (Top 125 students of 8,300 students in UIUC class)